EE1100: Basic Electrical Engineering
Tutorial 5 on Diode Circuits and Applications

PART A

Q1. Describe briefly how p-n junction is formed.

Q2. Write down Poisson's equation of space charge density. From Poisson's equation, write expression for electric field and voltage as a function of distance, *x*, *from* the center of space charge region. Draw the diagram for space charge density, electric field, and voltage.

Q3. Explain VI characteristics of p-n junction diode. Write equation and draw graph relating voltage and current p-n junction diode.

Q4. Explain the terms: i) Forward bias, ii) Reverse bias, iii) Cut-in voltage, iv) reverse saturation current, v) Forward bias resistance and reverse bias resistance.

Q5. Write down important applications of diodes.

Q6. Draw the circuits for i) half wave rectifier without and with capacitor filter, ii) Full wave rectifier circuit without and with capacitor filter. Draw typical output voltage and current waveforms. Also draw the input voltage and current waveforms. Assume a resistive load and consider diodes as ideal.

Q7. Define the ripple, ripple factor (RF) and form factor (FF) for output waveform for diode rectifier. Calculate and compare their values for half wave rectifier and full wave rectifier without and with capacitor filter. Consider the input supply to the rectifier circuit is $v_s(t) = V_m \sin 314t$.

Q8. In a half-wave rectifier, it is desired that the ripple at the output should be 10% of the peak value of the input voltage, $v_s(t) = V_m \sin 314t$. What is the value of capacitor for resistive load R_2 = 1 kΩ. Consider R_1 resistance in series with diode very small value.

Q9. Consider a full-wave rectifier with 5% ripple in the output voltage to that of the peak input voltage, $v_s(t) = V_m \sin 314t$. Design the capacitor filter for resistive load R_2 = 1 kΩ.

PART B

Q1. Consider the circuit in Fig. 1. Assume practical diodes (with turn-on voltage of 0.7 V). Determine currents I_1, I_2 and I_{D2}.

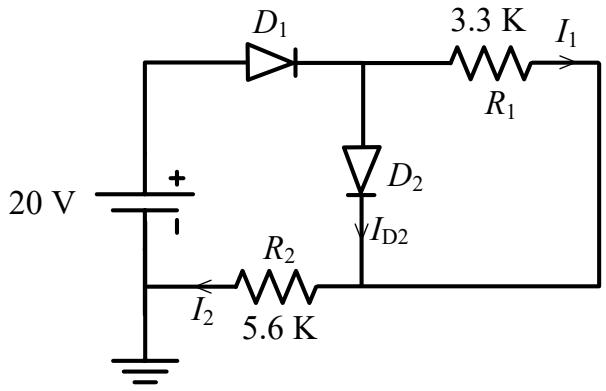


Fig. 1

[**Ans.**: I_1= 0.212 mA; I_2= 3.321 mA; I_{D2}= 3.109 mA]

Q2. Given the circuit of Fig. 2, find V_0. Assume practical diodes with a turn-on voltage of 0.7 V. Suppose inputs E_1, E_2 and E_3 are changed to 0, 0 and 10 V respectively, what is V_0.

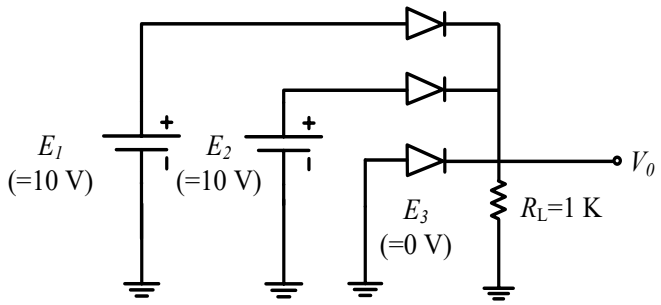


Fig. 2

[**Ans.**: V_0= 9.3 V]

Q3. Given the circuit and waveform in Fig. 3, sketch V_0. Assume diode is ideal.

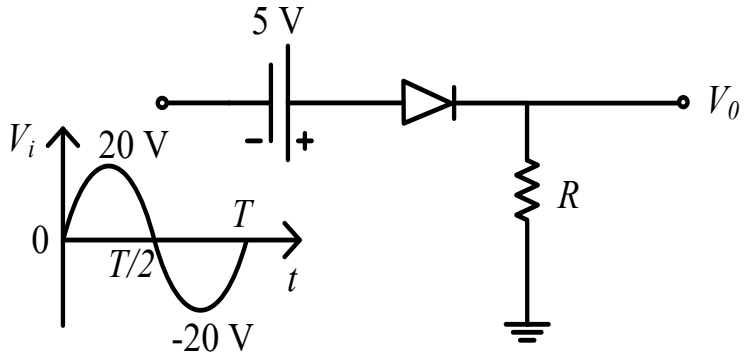


Fig. 3

Q4. Consider the circuit in Fig. 4 let inputs E_1, E_2 and E_3 be chosen as 0 V, 0 V and 0 V respectively. What is V_0? Suppose E_1 is changed to 10 V while the other two inputs remain 0 V, what will V_0 be? Consider other possible combinations of the three inputs ((0,10,0), (0,0,10), (10,0,10), (10,10,0), (10,10,10) etc.) and determine if the circuit is a realization of some digital logic element. What is the element? Assume practical diodes.

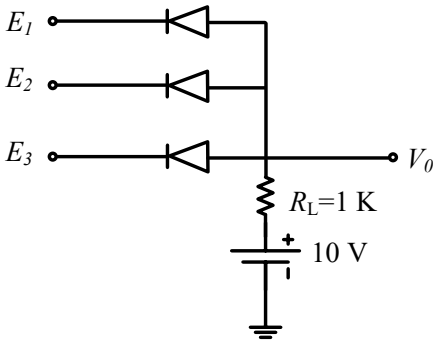


Fig. 4

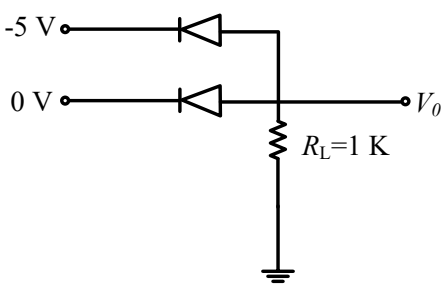


Fig. 5

[**Ans.**: $E_1 = E_2 = E_3$ =0 then V_0 = 0.7 V; $E_1 = E_2 = E_3$ = 10 V then V_0 = 10 V ; Circuit realizes AND gate]

Q5. What is V_0 in the circuit of Fig. 5? Assume diodes to be ideal.

[**Ans.**: $V_0 = -5$ V]

Q6. Given the circuit and waveform in Fig. 6, sketch V_0. Assume an ideal diode.

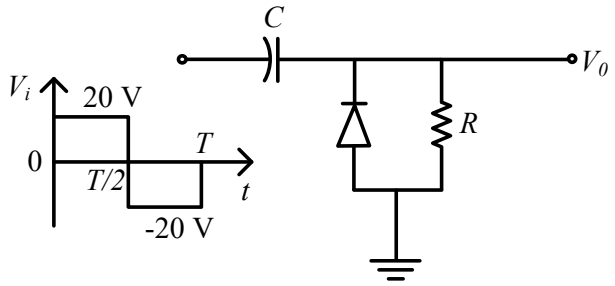


Fig. 6

Q7. Assume D1 and D_2 are ideal. $R_1 = R_2 = 100\ \Omega$. V_S is a square wave of period 1 ms.

Find current, i_L .

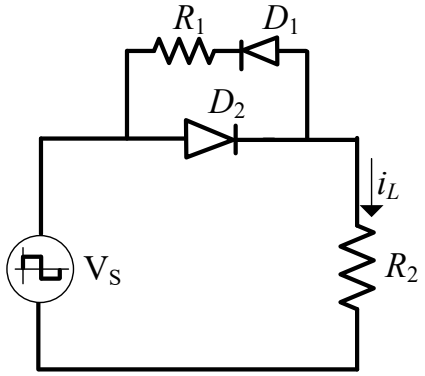


Fig. 7

[**Ans.**: During positive half cycle $i_L = 0.01\ V_S$; During negative half cycle $i_L = -0.005\ V_S$]

Q8. Draw the output waveform for the following circuit. The diode considered is ideal. R1= 10 kΩ, R2= 10 kΩ.

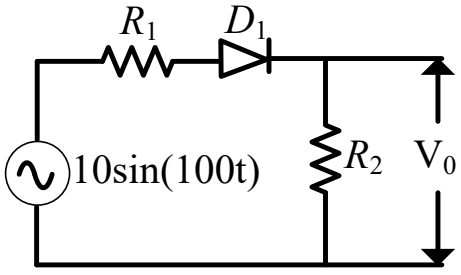


Fig. 8

[**Ans.**: During positive half cycle $V_0 = 5 \sin(100t)$ V; During negative half cycle $V_0 = 0$ V]

Q9. Consider the diode is ideal and plot the waveforms v0, current through battery and voltage across diode (V_D).

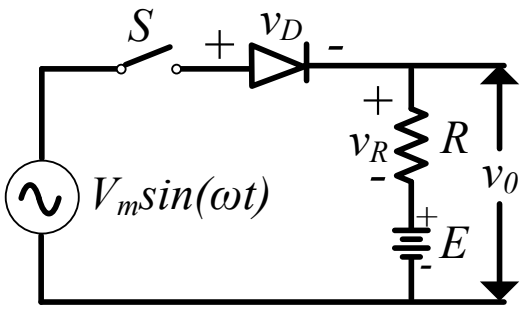
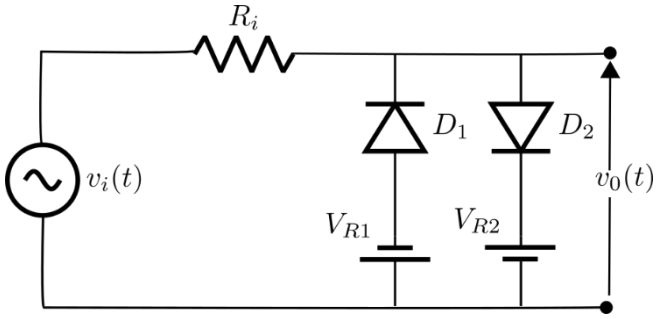


Fig. 9

Q10. In the circuit shown (diodes D_1, D_2 assumed ideal), the input is $v_i(t) = 10 \sin \omega t$ V with f = 1 kHz, R_i = 1 kΩ, and the reference voltages are $V_{R1} = -4$ V and $V_{R2} = +6$ V. Answer the following.

(a) State the condition on v_i (in terms of V_{R1} and V_{R2}) under which neither diode conducts, and give v_o in that region.
(b) Find the value of ωt (in the range 0 to 2π) at which D_1 first turns on, and the value at which it turns off. What is v_o throughout this interval?
(c) Find the value of ωt at which D_2 turns on and off, and give v_o throughout that interval.
(d) Sketch $v_o(t)$ over one complete cycle of $v_i(t)$, showing clearly the sinusoidal segment and the two flat (clipped) segments, with the transition points marked on the ωt axis.
(e) If each diode instead has a cut-in voltage $V_\gamma = 0.7$ V (rather than being ideal), find the corrected clipping levels and the revised switching points for D_1 and D_2.



[Ans.: [Ans.: (a) −4 V < v_i < 6 V ⟹ neither diode conducts, $v_o = v_i$; (b) D_1 ON at ωt = 203.58° (3.5531 rad), OFF at ωt = 336.42° (5.8717 rad), v_o = −4 V throughout; (c) D_2 ON at ωt = 36.87° (0.6435 rad), OFF at ωt = 143.13° (2.4981 rad), v_o = +6 V throughout; (d) v_o(t) follows the sine from 0° to 36.87° (rising 0→6 V), flat at +6 V from 36.87° to 143.13°, follows the sine from 143.13° to 203.58° (falling 6→−4 V), flat at −4 V from 203.58° to 336.42°, then follows the sine from 336.42° to 360° (rising −4→0 V)]

Q11. In the circuit shown in Fig. 10 each diode is assumed to have a resistance of 10 Ω. With respect to ground the voltages e1and e2 are equal and opposite. Specifically, e1= 100 sin ωt.

(a) Sketch the variation of the battery current as a function of time.

(b) Find the average value of this current.

[Ans.: (b) Average value of current = 0.0214 A]

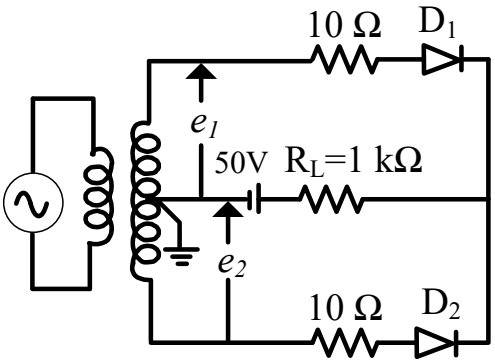


Fig. 10

Q12. In the circuit of Fig. 11 determine the magnitude of the maximum voltage appearing across R if the voltage source is e = Vm sin ωt. Assume that the resistance R has a very high value.

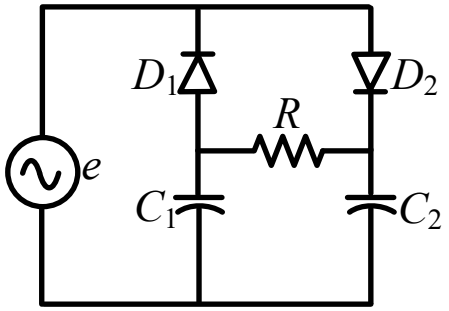


Fig. 11

[**Ans.**: $V_R = 2\ V_m$]